

Intesar Alam

Intesar Alam | LinkedIn

Email: intesaralam1113@gmail.com
Mobile: +92-313-9842499
Islamabad, Punjab - open to relocation

Mechanical Engineering Undergraduate | Final Year

Mechanical & Plant Engineering

EDUCATION

Ghulam Ishaq Khan Institute of Engineering Sciences & Technology, Topi	2022 - 2026
- Bachelor of Science – Mechanical Engineering	

PROFESSIONAL EXPERIENCE

IT MART- Peshawar	Jan 2020 – May2026
<i>Supply Chain Manager (Part-Time)</i>	

- Led demand forecasting and inventory planning for a retail store with ~**2,000+ SKUs**, reducing stockouts by **9.5%** and cutting excess inventory by **6%** through data-driven reorder point optimization and seasonal sales analysis.
- Managed end-to-end logistics and supplier relationships with **21+** vendors, negotiated pricing and delivery terms to lower procurement costs by **5.3%**.

Tarbela Power Station - (WAPDA)	Jul 2025 – Sep 2025
<i>Mechanical Intern</i>	

- Studied hydropower plant operations including turbines, governors, pumps, and auxiliary systems.
- Gained exposure to mechanical drawings, preventive maintenance, inspection routines, and industrial safety practices.

Oil & Gas Development Company Limited (OGDCL) – Tough Gas Field	Jun 2025 – Jul 2025
<i>Mechanical Intern</i>	

- Gained on-site exposure to gas production operations, rotating equipment, and maintenance workflows.
- Observed a full well intervention operation, gaining practical insight into the step-by-step procedure, equipment setup, safety protocols, and coordination required during live gas field activities.

LEADERSHIP & STUDENT EXPERIENCE

Team Invictus – AIAA Design/Build/Fly Competition	Dec 2022 – May 2024
<i>Sub-Head, Design & Documentation</i>	

- Contributed to design and aerodynamic analysis of a fixed-wing aircraft using SolidWorks and XFLR5; supported fabrication of competition-ready RC aircraft for an international contest with **100+** global teams.
- Co-led technical documentation for proposal submission, securing **worldwide 5th highest** proposal score.

IEEE GIKI Student Branch	Oct 2022 – Sep 2024
<i>Sub-Head, Liaison & Sponsorship</i>	

- Coordinated cross-functional communication for National Electronics Olympiad 2023 & 2024, engaging 20+ universities and **300+** participants.
- Secured **PKR 300,000+** in sponsorship funding through targeted marketing and corporate outreach initiatives.

ACADEMIC PROJECTS

Design and Development of a Vertical Axis Wind Turbine (Final Year Project) (Secured 2nd Position in the Faculty)
- Designed a helical VAWT for industrial exhaust energy recovery using CFD-based analysis and optimized rotor design. - Developed and tested a prototype, achieving an estimated 1.85 kW power output under operating conditions.

Industrial Conveyor System Design and Power Analysis
- Designed a 1.5 m conveyor (250 mm belt) with torque (2.99 Nm) and tension (113.3 N) calculations for a 100 N load - Selected a 75 W motor with 100 mm rollers (48 RPM) to optimize cost and manufacturability

ML-Based Predictive Maintenance Classification
- Built and trained 3-layer ANN (64–32 neurons) over 50 epochs on 1,500 samples, achieving 97.4% accuracy - Performed confusion matrix analysis (TN=1447, TP=14), attaining 87.5% precision in machine failure detection.

SKILLS / CERTIFICATIONS

Manufacturing & Operations: Process analysis, material handling systems, diagnostics, reliability concepts, HSE awareness
Engineering & Analysis Tools: AutoCAD SolidWorks ANSYS (CFX, FEA) MATLAB Simulink
Data & Analytical Skills: Basic machine learning, data analysis, FFT-based signal processing, performance evaluation
Certifications:
MATLAB Onramp (MathWorks) Simulink Onramp (MathWorks) Introduction to ANSYS Fluent (ANSYS)
The Bits and Bytes of Computer Networking (Google)